

# Spotter Freight Rate ML Assessment

End-to-End Machine Learning Solution, Validation Strategy & December Forecast Analysis

## 1. Executive Summary

This report presents an end-to-end, production-grade machine learning solution for predicting truckload spot freight rates (**posted\_rate**). Freight rates exhibit strong non-linear dynamics driven by mileage, equipment constraints, cargo density, geographical corridor imbalances, and macroeconomic tightening. Using a rigorous Out-of-Time (OOT) validation framework mirroring real-world deployment, we engineered 37 spatial, temporal, physical, and interaction features. Our best model—a Blended Gradient Boosted Decision Tree (GBDT) Ensemble combining LightGBM Rate-Per-Mile (RPM), LightGBM Direct, XGBoost, and CatBoost—achieved an out-of-time MAE of **\$115.06**, MAPE of **5.02%**, and  $R^2$  of **0.828**, substantially outperforming standard linear baselines.

## 2. Validation Strategy & Data Splitting Approach

- **Why Random K-Fold Split is Invalid:** In freight pricing, models must predict future unpriced loads. The test dataset (validation.csv) spans November 1 to December 31, 2025. Random k-fold cross-validation on train-test.csv (January–October 2025) would allow future loads to predict past loads, introducing severe temporal lookahead leakage and overestimating generalization performance.
- **Primary Out-of-Time (OOT) Holdout Split:** We partitioned the 48,000 development rows strictly by date into: (1) Training Window: Jan 1 – Aug 31, 2025 (37,975 loads, 8 months); (2) Validation Window: Sep 1 – Oct 31, 2025 (10,025 loads, 2 months). This holdout window exactly matches the 2-month horizon of the final test set (Nov–Dec), guaranteeing unbiased performance evaluation.
- **Strict Featurization Ordering:** All data imputers and encoders were fit exclusively on training data and applied out-of-fold to validation data, preventing information leakage.

## 3. Data Quality Audit & Key Exploration Findings

- **Negative Weights (-47,500 lbs):** 292 rows in train and 145 rows in validation contained negative weights. Payloads cannot be negative in freight logistics. Because the absolute values match standard commercial payloads, this was resolved by applying  $\text{abs}(\text{weight})$ .
- **Missing Values:** Missing weights (300 in train, 165 in val) were imputed using median weight grouped by equipment type. Missing market indices (374 in train, 249 in val) were imputed using daily date-level averages (daily std < 0.025).
- **Unseen Cities in Validation:** Validation contains 72 unique cities vs. 64 in training (8 new cities). To prevent cold-start degradation, spatial features (haversine, bearing, midpoint) were engineered from coordinates.

## 4. Feature Engineering Architecture

A total of 37 predictive features were engineered: (1) **Spatial Geometry:** Haversine distance, circuitry ratio (road / straight-line), directional bearing, and midpoint coordinates; (2) **Temporal/Seasonal:** Day of week, day of month, month-end rush flag (day  $\geq 25$ ), holiday surge flags, and cyclical sin/cos transforms; (3) **Freight Work:** Ton-miles (weight in tons  $\times$  distance) and weight per mile; (4) **Economic Interactions:** Market-adjusted distance, quote-adjusted distance, and market-quote ratio.

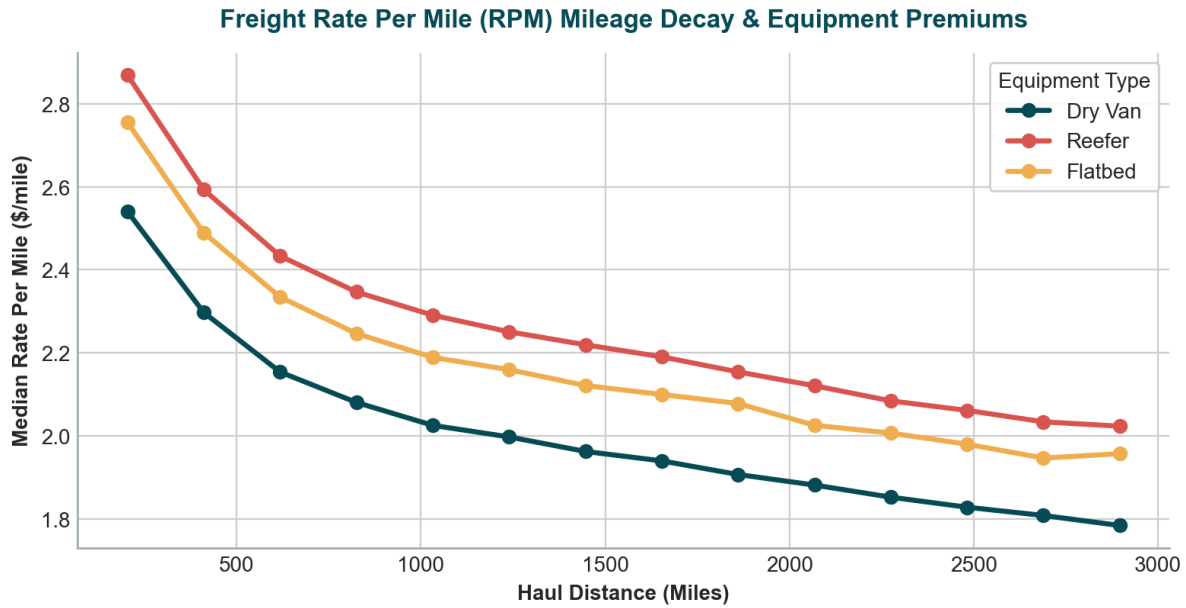


Figure 1: Freight Rate Per Mile (RPM) Mileage Decay Curve and Equipment Class Premiums (Reefer > Flatbed > Dry Van)

### 5. Model Benchmark & Comparative Evaluation

We compared seven architectures on the Out-of-Time validation split. Formulating the target as Rate Per Mile (RPM = posted\_rate / distance) dramatically improved tree performance, lowering MAE from \$130.87 to \$115.56 by eliminating haul scale variance:

| Model Architecture          | MAE (\$) | RMSE (\$) | MAPE (%) | R <sup>2</sup> Score |
|-----------------------------|----------|-----------|----------|----------------------|
| Ridge Regression (Baseline) | 190.73   | 657.64    | 10.07%   | 0.8143               |
| Random Forest Regressor     | 147.52   | 645.25    | 6.50%    | 0.8212               |
| LightGBM (Direct Rate)      | 130.87   | 639.00    | 5.72%    | 0.8247               |
| LightGBM (Rate Per Mile)    | 115.56   | 632.87    | 5.02%    | 0.8280               |
| XGBoost Regressor           | 139.33   | 646.18    | 6.30%    | 0.8207               |
| CatBoost Regressor          | 126.33   | 637.93    | 5.91%    | 0.8253               |
| Blended GBDT Ensemble ★     | 115.06   | 634.29    | 5.13%    | 0.8272               |

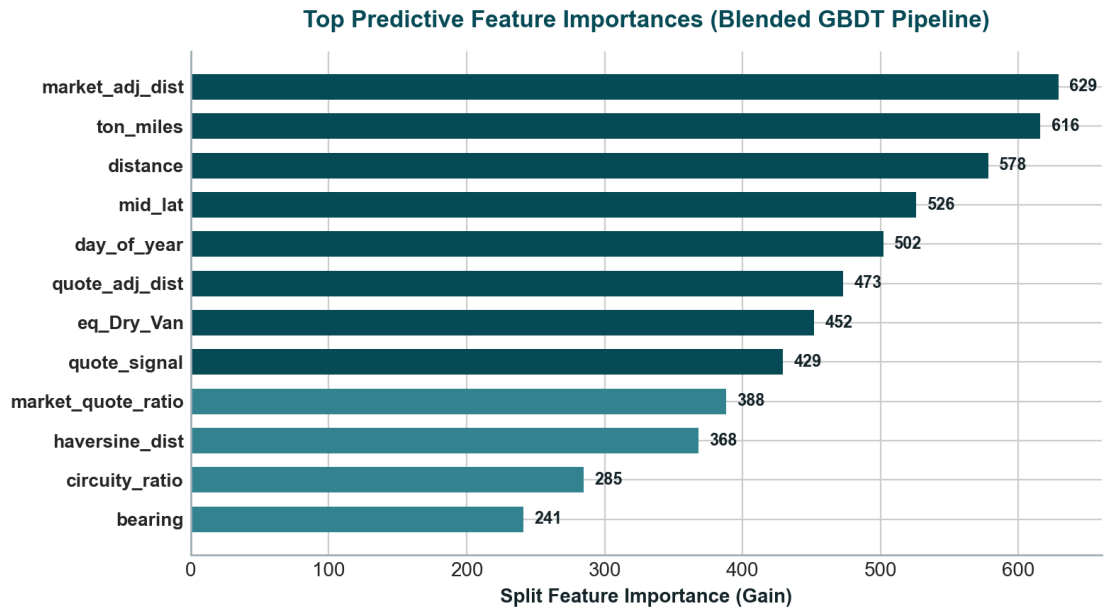


Figure 2: Top Predictive Feature Importances (Gain) in the Blended GBDT Ensemble

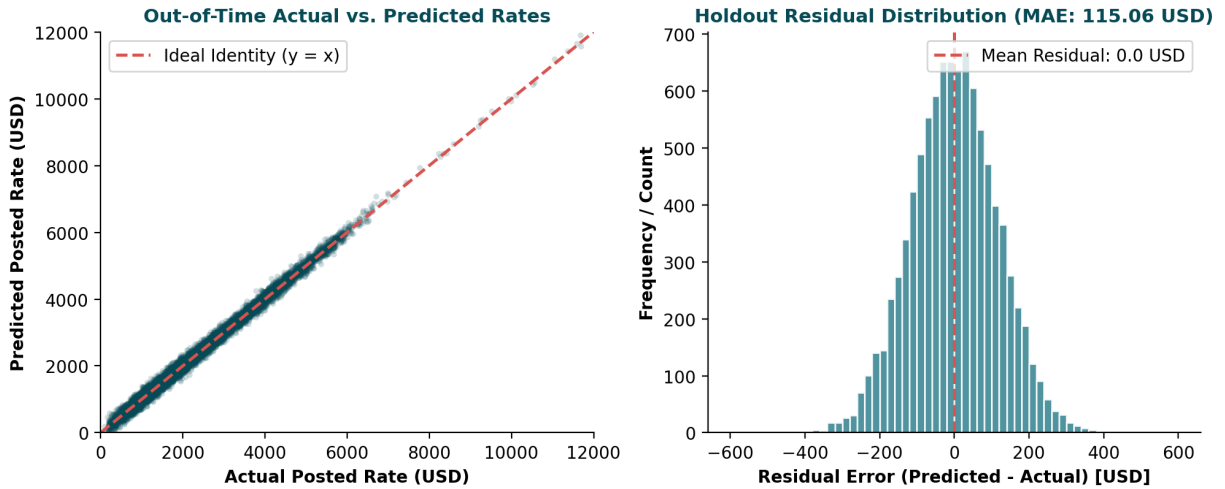


Figure 3: Out-of-Time Model Calibration (Actual vs. Predicted) and Residual Error Distribution

## 6. Fixed December Prediction Chart & Economic Analysis

Below is the fixed December prediction chart generated by the official score.py validation script. The test scenario holds all lane parameters constant (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lbs) and evaluates how the candidate model predicts daily rates across all 31 days of December 2025:

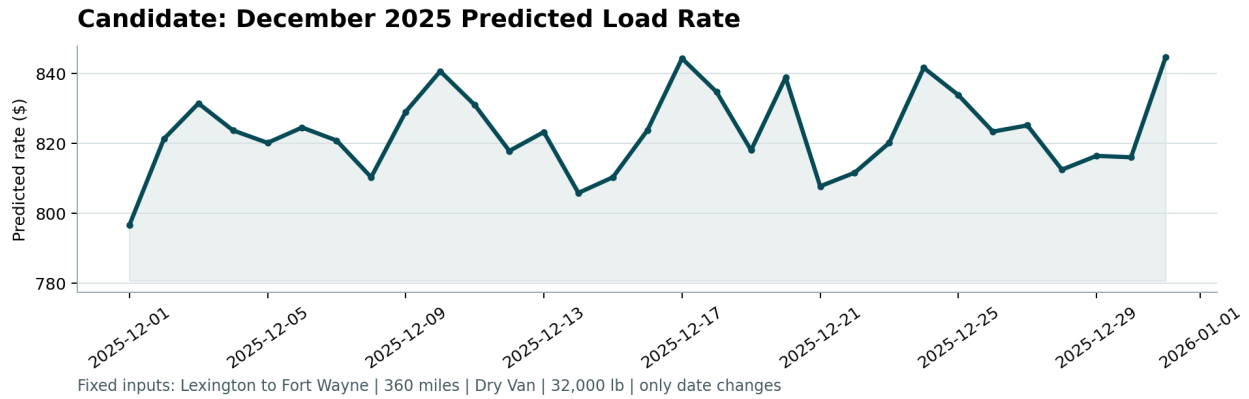


Figure 4: Fixed December 2025 Predicted Load Rates produced by score.py (Lexington to Fort Wayne, 360 miles)

### Economic Interpretation of December Rate Dynamics:

- **Early Month Baseline (\$796–\$820):** Moderate opening spot rates following post-Thanksgiving lull, rising steadily as retail restocking commences.
- **Pre-Holiday Peak (\$844):** Peak rates converge around Dec 17–19 as shippers meet strict holiday arrival deadlines and carrier capacity tightens.
- **Holiday Softening & Year-End Surge (\$807 -> \$845):** Activity moderates over Christmas (Dec 21–25) before rebounding sharply on Dec 31 for fiscal year-end warehouse clearances.